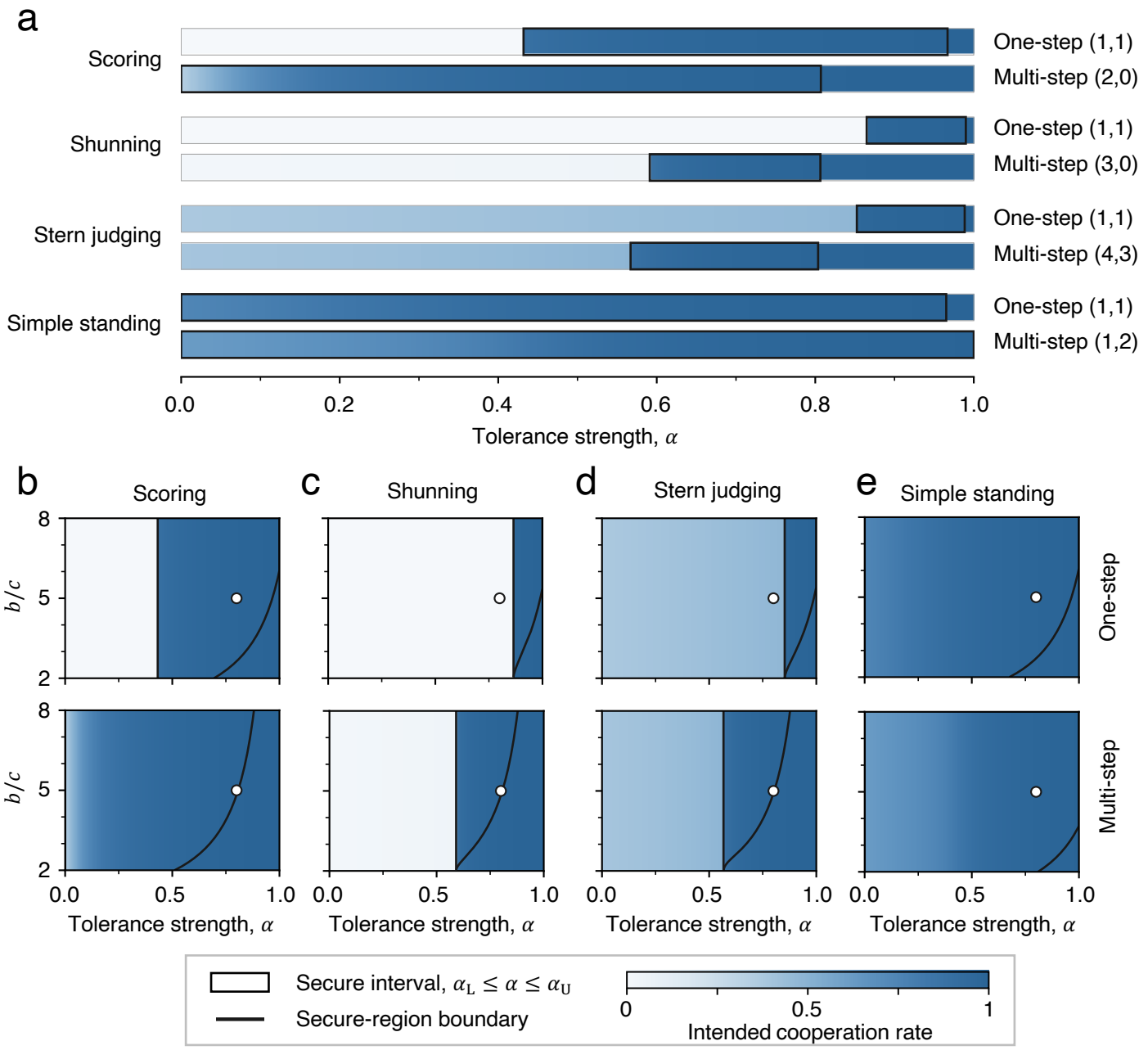

a
Scoring
Shunning
Stern judging
Simple standing
One-step (1,1)
Multi-step (2,0)
One-step (1,1)
Multi-step (3,0)
One-step (1,1)
Multi-step (4,3)
One-step (1,1)
Multi-step (1,2)
0.0
0.2
0.4
0.6
0.8
1.0
Tolerance strength, α
b
Scoring
c
Shunning
d
Stern judging
e
Simple standing
b/c
8
5
2
One-step
Multi-step
0.0
0.5
1.0
Tolerance strength, α
Secure interval, α_L ≤ α ≤ α_U
Secure-region boundary
0
0.5
1
Intended cooperation rate